Preface

Notes and Stuff

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Chapter 1
Topological Vector Spaces

Throughout, assume $\mathbf{F}$ to be either $\mathbf{R}$ or $\mathbf{C}$.

§ 1.1. Basic Definitions and Examples

I. Vector Topologies

Definition 1.1.1. Suppose T is a topology on an $\mathbf{F}$-vector space X such that:

- for every $x \in X$, the set $\{x\}$ is closed;
- the vector space operations are continuous with respect to T.

Under these conditions, T is called a *vector topology* and X is said to be a *topological vector space* (abbrev. *TVS*).

Definition 1.1.2. Let X be a TVS.

(a) A set $C \subset X$ is said to be *convex* if $tC + (1-t)C \subset C$ for every $0 \leqslant t \leqslant 1$.

(b) A set $B \subset X$ is said to be *balanced* if $\alpha B \subset B$ for every $\alpha \in \mathbf{F}$ with $|\alpha| \leqslant 1$

Proposition 1.1.1. *Let X be a TVS.*

(a) The translation operator $T_a : X \to X$ given by $T_a = a + x$ is a homeomorphism.

(b) The multiplication operator $M_\lambda : X \to X$ given by $M_\lambda(x) = \lambda x$ is a homeomorphism.

Proof. The continuity and bijectivity of T_a and M_λ follow from the vector space axioms. Their inverses, T_{-a} and $M_{1/\lambda}$, are also continuous due to the vector space axioms. ■

Remark. Recall that:

- a *neighborhood* of a point $x \in X$ is any open set that contains x;
- a collection N of neighborhoods of a point $x \in X$ is a *local base at x* if every neighborhood of x contains a member of N.

A consequence of Proposition 1.1.1 is that every vector topology is *translation-invariant*; i.e., a set $E \subset X$ is open if and only if each of its translates $a + E$ is open. As such, any vector topology is completely determined by any local base, which leads us to the following definition.

Definition 1.1.3. A *local base* of a TVS is a collection $\mathscr{B}$ of neighborhoods of 0 such that every neighborhood of 0 contains a member of $\mathscr{B}$.

II. Separation Properties

Lemma 1.1.2. *Let X be a TVS. If W is a neighborhood of 0 in X, then there is a neighborhood U of 0 such that $U = -U$ and $U + U \subset W$.*

Proof. Let $P : X \times X \to X$ denote addition. Since addition is continuous, it is continuous at 0. Hence $P^{-1}(W)$ is a neighborhood of $(0,0)$. In particular, there exists open sets V_1 and V_2 of X such that $(0,0) \in V_1 \times V_2 \subseteq P^{-1}(W)$. Now define $U := V_1 \cap V_2 \cap (-V_1) \cap (-V_2)$. This is an open set of X containing zero which is symmetric and contained in both V_1 and V_2. Thus:

$$U + U = P(U \times U) \subseteq P(V_1 \times V_2) \subseteq W. \qquad \blacksquare$$

Lemma 1.1.3. *Let X be a TVS. Suppose K and C are subsets of X where K is compact, C is closed, and $K \cap C = \emptyset$. Then 0 has a neighborhood V such that:*

$$(K + V) \cap (C + V) = \emptyset.$$

Proof. Suppose $x \in K$. Then $x \notin C$, implying that C^c is a neighborhood of x. By Proposition 1.1.1, the translation $C^c - x$ is a neighborhood of 0, so by Lemma 1.1.2 we can find a symmetric neighborhood V_x of 0 such that:

$$V_x + V_x + V_x \subset C^c - x.$$

This can be rewritten as:

$$x + V_x + V_x + V_x \subset C^c.$$

Claim: $(x + V_x + V_x) \cap (V_x + C) = \emptyset$. Indeed, if $z \in (x + V_x + V_x) \cap (V_x + C)$, then $z = x + v_1 + v_2$ and $z = v_3 + c$ for $v_1, v_2, v_3 \in V_x$ and $c \in C$. Equating the two yields $x + v_1 + v_2 - v_3 = c$, which is a contradiction as we showed $(x + V_x + V_x + V_x) \cap C = \emptyset$.

Since $x + V_x$ is a neighborhood of x, we can cover K as follows:

$$K \subset \bigcup_{x \in K} (x + V_x).$$

Since K is compact, there are finitely many points $x_1, \ldots, x_n$ in K such that:

$$K \subset (x_1 = V_{x_1}) \cup \ldots \cup (x_n + V_{x_n}).$$

Define $V := V_{x_1} \cap \ldots \cap V_{x_n}$. Then:

$$K + V \subset \bigcup_{i=1}^{n} (x_i + V_{x_i} = V) \subset \bigcup_{i=1}^{n} (x_i + V_{x_i} + V_{x_i}).$$

By our claim, no term in the last union intersects $C + V$, establishing the proof. $\blacksquare$

Theorem 1.1.4. *Let X be a TVS. Suppose K and C are subsets of X where K is compact, C is closed, and $K \cap C = \emptyset$. There exists disjoint open sets that contain K and C, respectively.*

Proof. From Lemma 1.1.3, note that $K + V$ is a union of translates $x + V$ of V. Thus $K + V$ is an open set that contains K. Similarly, as C is the union of translates $x + C$, it is an open set containing C. $\blacksquare$

Corollary 1.1.5. *If X is a TVS, and $\mathscr{B}$ is a local base for X, then every member of $\mathscr{B}$ contains the closure of some member of $\mathscr{B}$.*

Proof. Let $B \in \mathscr{B}$. Since $\{0\}$ is compact and $X \setminus B$ is closed, Lemma 1.1.3 says there exists a neighborhood V of 0 such that:

$$(\{0\} + V) \cap (X \setminus B) = \emptyset.$$

This simpifies to:

$$V \cap (X \setminus B) = \emptyset.$$

It is a general fact that an open set which is disjoint from a set is also disjoint from its closure. Hence:

$$\overline{V} \cap (X \setminus B) = \emptyset.$$

This implies:

$$\overline{V} \subset B.$$

Since V is a neighborhood of 0 and $\mathscr{B}$ is a local base, we can find a $B' \in \mathscr{B}$ such that $B' \subset V$. Hence $\overline{B'} \subset \overline{V} \subset B$, establishing the claim. ∎

Corollary 1.1.6. *Every TVS is a Hausdorff space.*

Proof. Apply Lemma 1.1.3 to a pair of distinct points in the place of K and C. ∎

III. Local Convexity

Definition 1.1.4. Let X be a TVS and T its vector topology.

(a) X is *locally convex* if there is a local base $\mathscr{B}$ whose members are convex.

(b) X is a *Fréchet space* if X is locally convex and its topology T is induced by a complete invariant metric.

Definition 1.1.5. A *seminorm* on an $\mathbf{F}$-vector space is a function $p : X \to \mathbf{R}$ such that:

- $p(x+y) \leqslant p(x) + p(y)$;
- $p(\alpha x) = |\alpha| p(x)$

for all $x, y \in X$ and $\alpha \in \mathbf{F}$.

Proposition 1.1.7. *Suppose p is a seminorm on an $\mathbf{F}$-vector space X.*

(a) $p(0) = 0$.

(b) $|p(x) - p(y)| \leqslant p(x-y)$.

(c) $p(x) \geqslant 0$.

(d) $\{x : p(x) = 0\}$ *is a subspace of* X.

Proof. ∎

Definition 1.1.6. A family $\mathscr{P}$ of seminorms on an $\mathbf{F}$-vector space X is called *separating* if each $x \neq 0$ corresponds to at least one $p \in \mathscr{P}$ with $p(x) \neq \emptyset$.

Definition 1.1.7. Let X be a $\mathbf{F}$-vector space.

(a) A convex set $A \subset X$ is *absorbing* if for every $x \in X$, there exists a scalar $t > 0$ (possibly depending on x) such that $x \in tA$.

(b) If A is an absorbing set, the *Minkowski functional* of A is a function $\mu_A : X \to \mathbf{F}$ defined by:

$$\mu_A(x) := \inf\{t : t^{-1}x \in A\}$$

Proposition 1.1.8. *Suppose p is a seminorm on an $\mathbf{F}$-vector space X. The set $B := \{x : p(x) < 1\}$ is convex, balanced, absorbing, and $p = \mu_b$.*

Proof. For any $x \in X$ and $\alpha \in \mathbf{F}$ we have:

$$\begin{aligned} p(\alpha x) &= |\alpha| p(x) \\ &\leqslant p(x) \\ &< 1. \end{aligned}$$

Thus B is balanced. For any $x, y \in X$ and $0 < t < 1$ we have:

$$\begin{aligned} p(tx + (1-t)y) &\leqslant p(tx) + p((1-t)y) \\ &= tp(x) + (1-t)p(y) \\ &\leqslant t + (1-t) \\ &= 1. \end{aligned}$$

Thus B is convex. If $x \in X$ and $s > p(x)$, we have:

$$\begin{aligned} p(s^{-1}x) &= s^{-1}p(x) \\ &< 1. \end{aligned}$$

Thus $s^{-1}x \in B$, hence B is absorbing. We will now show that $p = \mu_B$. Let $x \in X$ and $s \in \mathbf{F}$ such that $p(x) < s$. Then $p(s^{-1}x) < 1$, implying that $s \in \{t : t^{-1}x \in B\}$. This means $\mu_B(x) \leqslant s$, and taking the infimum over all s such that $p(x) < s$ gives $\mu_B(x) \leqslant p(x)$. Conversely, let $x \in X$ and $r \in \mathbf{F}$ such that $0 < r \leqslant p(x)$. Then $1 \leqslant p(r^{-1}x)$, implying that $r \notin \{t : t^{-1}x \in B\}$. We've established the inclusion $\{t : t^{-1}x \in B\} \subseteq (p(x), \infty)$, and taking the infimum of both sides gives $p(x) \leqslant \mu_B(x)$. Thus $p = \mu_B$. ■

Theorem 1.1.9. *Suppose A is a convex absorbing set in a vector space X.*

(a) $\mu_A(x + y) \leqslant \mu_A(x) + \mu_A(y)$.

(b) $\mu_A(tx) = t\mu_A(x)$ *if* $t \geqslant 0$.

(c) μ_A *is a seminorm if A is balanced.*

(d) If $B = \{x : \mu_A(x) < 1\}$ and $C = \{x : \mu_A(x) \leqslant 1\}$, then $B \subset A \subset C$ and $\mu_B = \mu_A = \mu_C$.

Proof. ■

Theorem 1.1.10. *Let X be a TVS. Suppose $\mathscr{B}$ is a convex balanced local base in X.*

(a) For every $V \in \mathscr{B}$, we have $V = \{x \in X : \mu_V(x) < 1\}$.

(b) $\{\mu_V : V \in \mathscr{B}\}$ is a separating family of continuous seminorms.

Proof. ■

Theorem 1.1.11. *Suppose $\mathscr{P}$ is a separating family of seminorms on a vector space X. Associate to each $p \in \mathscr{P}$ and to each $n \in \mathbf{N}$ the set:*

$$V(p,n) := \left\{x : p(x) < \frac{1}{n}\right\}.$$

Let $\mathscr{B}$ be the collection of all finite intersections of the sets $V(p,n)$. Then $\mathcal{B}$ is a convex balanced local base for a topology T on X such that:

- *every $p \in \mathscr{P}$ is continuous;*
- *a set $E \subset X$ is bounded if and only if every $p \in \mathscr{P}$ is bounded on E.*

Proof. ■

Theorem 1.1.12. *Let X be a TVS and $\mathscr{B}$ a convex balanced local base in X. Let T_1 denote the topology generated by $\mathscr{B}$. Let T_2 denote the topology generated by $\{V(\mu_V, n) : V \in \mathscr{B}, n \in \mathbf{N}\}$, where $V(\mu_V, n)$ is defined as per Theorem 1.1.11. Then $T_1 = T_2$.*

Proof. ■

§ 1.2. Weak Topologies

I. Preliminaries

Definition 1.2.1. Let T_1 and T_2 be two topologies on a set X. If $T_1 \subset T_2$ (that is, every set open in T_1 is open in T_2), then we say T_1 is <u>*weaker*</u> than T_2, or that T_2 is <u>*stronger*</u> than T_1.

Proposition 1.2.1. *Let T_1 and T_2 be topologies on a set X. Suppose $T_1 \subset T_2$, T_1 is Hausdorff, and T_2 is compact. Then $T_1 = T_2$.*

Proof. Let $F \subset X$ be T_2-closed. Since X is T_2-compact, and since every closed subset of a compact set is compact, we have that F is T_2-compact. Since $T_1 \subset T_2$, every T_1-open cover of F is also a T_2-open cover. Hence F is T_1-compact. Since T_1 is Hausdorff, every compact set must be closed; i.e., F is T_1-closed. This establishes the inclusion $T_2 \subset T_1$, completing claim. ■

Definition 1.2.2. Let X be a set and $\mathscr{F}$ is a nonempty family of functions $f : X \to Y_f$, where each Y_f is a topological space. Let T be the collection of all unions of finite intersections of sets $f^{-1}(V)$, with $f \in \mathscr{F}$ and $V \in Y_f$. Then T is a topology on X, and is called the *weak topology on X induced by $\mathscr{F}$*, or more concisely, the *$\mathscr{F}$-topology* of X.

Proposition 1.2.2. *If $\mathscr{F}$ is a family of mapping $f : X \to Y_f$, where X is a set and each Y_f is a Hausdorff space, and if $\mathscr{F}$ separates points on X, then the $\mathscr{F}$-topology of X is a Hausdorff topology.*

Proof. Let $p, q \in X$ with $p \neq q$. Since $\mathscr{F}$ separates points on X, ; ■

Chapter 2
Weak Topologies

Let X be a LCS. For any $x \in X$, recall that the *evaluation map* is $\hat{\cdot} : X^* \to \mathbf{R}$ given by $\hat{x}(f) := f(x)$. We introduce the following notation for convenience.

- For $f \in X^*$, define $\langle \cdot, f \rangle : X \to \mathbf{F}$ by $x \mapsto f(x)$.
- For $x \in X$, define $\langle x, \cdot \rangle : X^* \to \mathbf{F}$ by $f \mapsto \hat{x}(f) = f(x)$.

We call $\langle \cdot, \cdot \rangle$ the *canonical evaluation map*.

§ 2.1. Basic Definitions and Examples

Definition 2.1.1. Let X be a LCS.

(a) The *weak topology* on X, denoted $\sigma(X, X^*)$ (or wk), is the topology generated by the family of seminorms $\{p_f : X \to \mathbf{F} : f \in X^*\}$, where $p_f(x) := |\langle x, f \rangle|$.

(b) *weak-* topology* on X^*, denote $\sigma(X^*, X)$ (or $wk*$), is the topology generated by the family of seminorms $\{p_x : X^* \to \mathbf{F} : x \in X\}$, where $p_x(f) := |\langle x, f \rangle|$.

Remark. Note that $\sigma(X, X^*)$ has $\{p_f^{-1}(U) : f \in X^*, U \subseteq \mathbf{F} \text{ open }\}$ as a subbase. As such, given $f_1, \ldots, f_k \in X^*$ and $U_1, \ldots U_k$ open in $\mathbf{F}$, an arbitrary basis element is of the form $\bigcap_{i=1}^k f_k^{-1}(U_k)$.

Proposition 2.1.1. *Let X be a LCS with the weak topology. Then U is open in X if and only if for every $x_0 \in U$ there exists $\epsilon > 0$ and $f_1, \ldots, f_n \in X^*$ such that:*

$$\bigcap_{k=1}^{n} \{x \in X : |\langle x - x_0, f_k \rangle| < \epsilon\} \subseteq U.$$

Chapter 3
Banach Algebras

Unless otherwise stated, let $\mathbf{F} = \mathbf{C}$.

§ 3.1. Definitions and Examples

Definition 3.1.1. An *algebra* is an $\mathbf{F}$-vector space V equipped with a multiplication operator making V into a ring such that, for all $\alpha \in \mathbf{F}$ and $x, y \in V$, we have:

- $\alpha(xy) = (\alpha x)y = x(\alpha y)$.

Definition 3.1.2. A *normed algebra* is an algebra V equipped with a norm $\|\cdot\|$ making V into a normed linear space such that, for all $x, y \in V$, we have:

- $\|xy\| \leqslant \|x\| \|y\|$.

If V has an identity e, then it is assumed that $\|e\| = 1$.

Proposition 3.1.1. *If V*

Definition 3.1.3. An $\mathbf{F}$-*algebra* is an $\mathbf{F}$-vector space A equipped with a multiplication operation that makes V into a ring such that for all $\alpha \in \mathbf{F}$ and $a, b \in V$, we have $\alpha(ab) = (\alpha a)b = a(\alpha b)$. An element $e \in A$ satisfying $ea = ae = a$ for all $a \in A$ is called an *identity*.

Proposition 3.1.2. *Let A be a normed $\mathbf{F}$-algebra with identity e. Suppose that, for every $\alpha \in \mathbf{F}$, we have $\|\alpha e\| = |\alpha|$. Then the map $\mathbf{F} \to A$ given by $\alpha \mapsto \alpha e$ is an isometric isomorphism of $\mathbf{F}$ onto the subalgebra $\mathbf{F}e = \{\alpha e : \alpha \in \mathbf{F}\}$.*

Remark. It will be assumed that $\mathbf{F} \subseteq A$ via the above identification. As such, the identity of an algebra will always be denoted by 1.

Definition 3.1.4. A *normed* $\mathbf{F}$-*algebra* is an $\mathbf{F}$-algebra A equipped with a norm $\|\cdot\|$ such that, for all $a, b \in A$, we have $\|ab\| \leqslant \|a\| \|b\|$. If A is a Banach space with respect to this norm, then A is called a *Banach algebra*.

Proposition 3.1.3. *Let A be a Banach algebra without identity with norm $\|\cdot\|$. Define binary operations on $A \times \mathbf{F}$ by:*

- $(a, \alpha) + (b, \beta) := (a + b, \alpha + \beta)$;
- $\beta(a, \alpha) := (\beta a, \beta\alpha)$;
- $(a, \alpha)(b, \beta) := (ab + \alpha b + \beta a, \alpha\beta)$.

Moreover, define:

- $\|(a,\alpha)\| := \|a\| + |\alpha|$.

Then $A \times \mathbf{F}$ *with the above binary operations and norm is a Banach algebra with identity* $(0,1)$. *Furthermore,* $a \mapsto (a,0)$ *is an isometric isomorphism*

Proof. ■

Definition 3.1.5. Let A be an $\mathbf{F}$-algebra. *left ideal* (resp. *right ideal*) of A is a subalgebra M of A such that $ax \in M$ (resp. $xa \in M$) whenever $a \in A$ and $x \in M$. If A is both a left and right ideal, we simply refer to it as an *ideal*.

We drop the asdf and refer to it as an ideal.

§ 3.2. The Spectrum